

Windows Event Log Monitoring Lab using Sysmon

1. Introduction

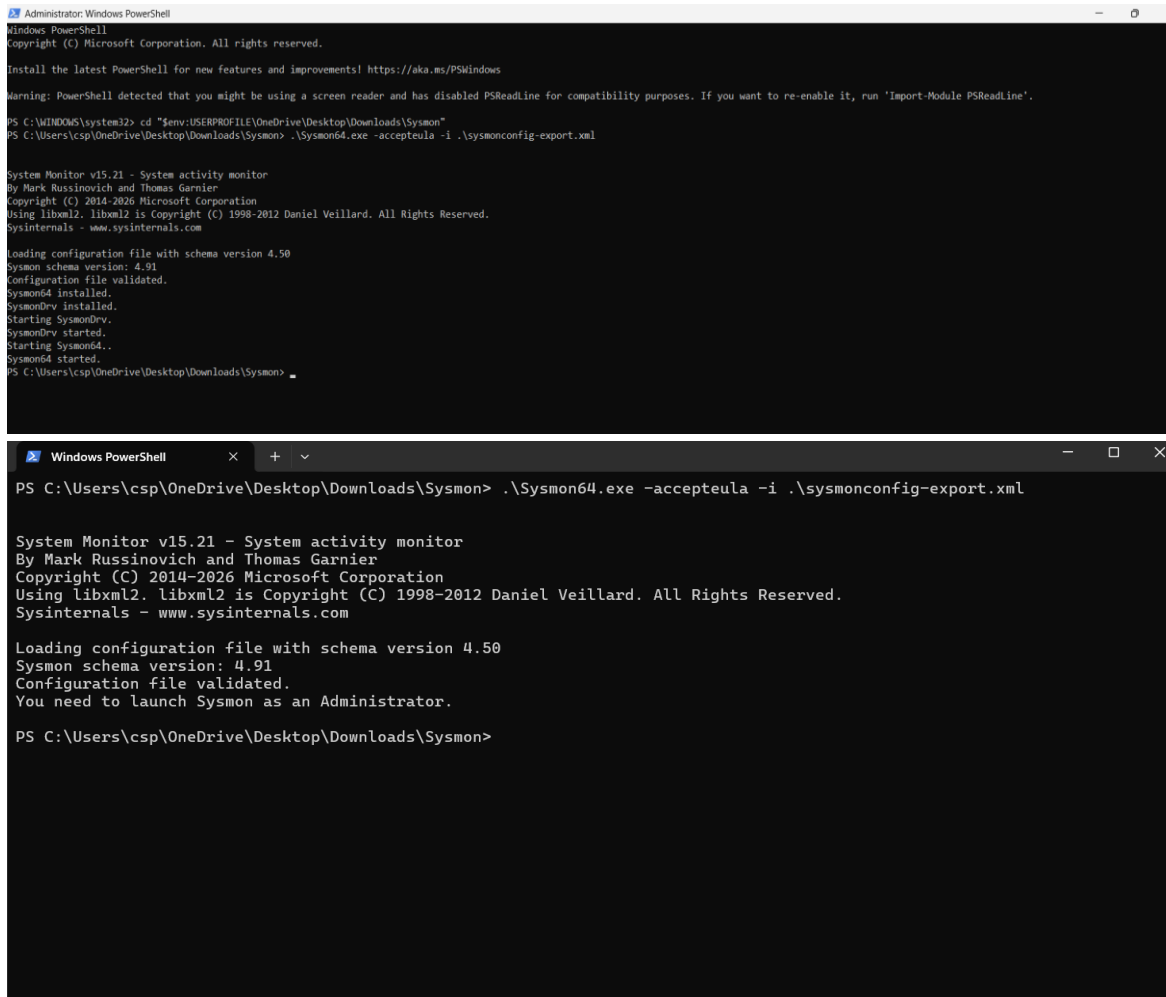
This project demonstrates Windows event monitoring using Sysmon and Event Viewer. The objective was to collect and analyze Windows security events commonly investigated by SOC Analysts.

2. Tools Used

- Windows 11 Virtual Machine
- VMware Workstation
- Sysmon
- Event Viewer
- PowerShell

3. Sysmon Installation

Sysmon was installed and configured to generate detailed Windows telemetry logs. Event Viewer was used to investigate the generated events.



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

Warning: PowerShell detected that you might be using a screen reader and has disabled PSReadLine for compatibility purposes. If you want to re-enable it, run 'Import-Module PSReadLine'.

PS C:\WINDOWS\system32> cd "$env:USERPROFILE\OneDrive\Desktop\Downloads\Sysmon"
PS C:\Users\csp\OneDrive\Desktop\Downloads\Sysmon> .\Sysmon64.exe -accepteula -i .\sysmonconfig-export.xml

System Monitor v15.21 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.91
Configuration file validated.
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
PS C:\Users\csp\OneDrive\Desktop\Downloads\Sysmon>

Windows PowerShell
PS C:\Users\csp\OneDrive\Desktop\Downloads\Sysmon> .\Sysmon64.exe -accepteula -i .\sysmonconfig-export.xml

System Monitor v15.21 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.91
Configuration file validated.
You need to launch Sysmon as an Administrator.
PS C:\Users\csp\OneDrive\Desktop\Downloads\Sysmon>
```

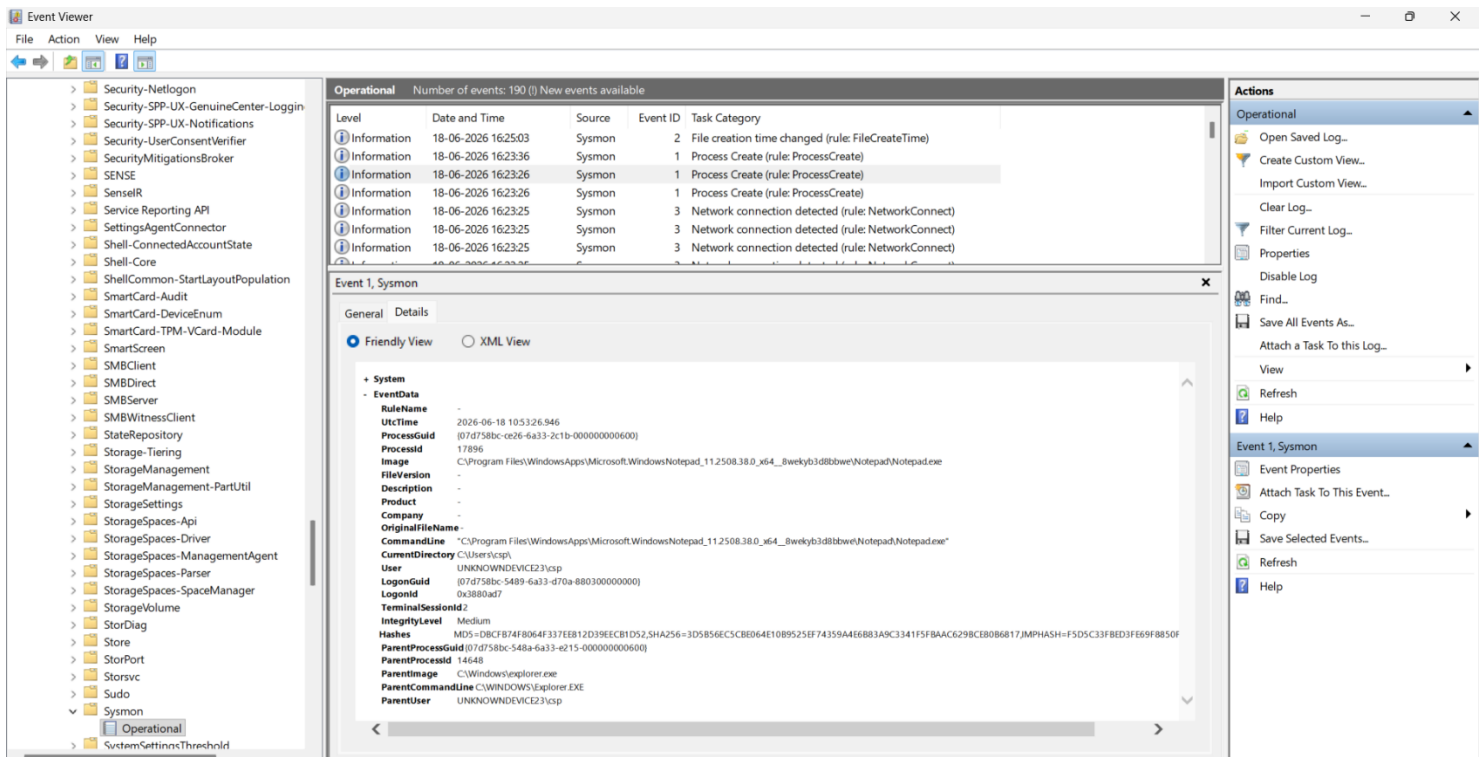
4. Event ID 1 - Process Creation

Activities performed:

- notepad.exe
- powershell.exe
- whoami.exe
- hostname.exe
- ipconfig.exe
- chrome.exe

Findings:

- Successfully monitored process creation events.
- Identified parent-child process relationships.
- Verified user activity through command-line execution logs.



EventID1-Process Creation Notepad.exe

Event Viewer

File Action View Help

Security-Netlogon
Security-SPP-UX-GenuineCenter-Login
Security-SPP-UX-Notifications
Security-UserConsentVerifier
SecurityMitigationsBroker
SENSE
SenseR
Service Reporting API
SettingsAgentConnector
Shell-ConnectedAccountState
Shell-Core
ShellCommon-StartLayoutPopulation
SmartCard-Audit
SmartCard-DeviceEnum
SmartCard-TPM-VCARD-Module
SmartScreen
SMBClient
SMBDirect
SMBServer
SMBWitnessClient
StateRepository
Storage-Tiering
StorageManagement
StorageManagement-PartUtil
StorageSettings
StorageSpaces-API
StorageSpaces-Driver
StorageSpaces-ManagementAgent
StorageSpaces-Parser
StorageSpaces-SpaceManager
StorageVolume
StorDiag
Store
StorPort
Storsvc
Sudo
Sysmon
SystemSettingsThreshold

Operational Number of events: 359 (0) New events available

Level	Date and Time	Source	Event ID	Task Category
Information	18-06-2026 16:54:15	Sysmon	11	File created (rule: FileCreate)
Information	18-06-2026 16:54:15	Sysmon	1	Process Create (rule: ProcessCreate)
Information	18-06-2026 16:54:11	Sysmon	11	File created (rule: FileCreate)
Information	18-06-2026 16:54:10	Sysmon	1	Process Create (rule: ProcessCreate)
Information	18-06-2026 16:54:08	Sysmon	1	Process Create (rule: ProcessCreate)
Information	18-06-2026 16:54:08	Sysmon	1	Process Create (rule: ProcessCreate)

Event 1, Sysmon

General Details

Friendly View XML View

EventData

RuleName -

UtcTime 2026-06-18 11:24:15.071

ProcessGuid {07d758bc-d55f-6a33-8c1b-000000000600}

ProcessId 20772

Image C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe

FileVersion 10.0.26100.5074 (WinBuild.160101.0800)

Description Windows PowerShell

Product Microsoft® Windows® Operating System

Company Microsoft Corporation

OriginalFileName powershell.exe

CommandLine "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -ExecutionPolicy Bypass

CurrentDirectory C:\Windows\system32\

User UNKNOWN\DEVICE23\csp

LogonGuid {07d758bc-5489-6a33-7106-880300000000}

LogonId 0x3880671

TerminalSessionId 2

IntegrityLevel High

Hashes MD5=A97E6573B97B44C961228FA543A02EA1SHA256=0FF6F2C9ABC7E2833A5F7E16DE1622E5DBA70396F31C7D5F56381870317BFC46,IMPHASH=AFAC6C9034

ParentProcessGuid {07d758bc-d55a-6a33-891b-000000000600}

ParentProcessId 20444

ParentImage C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe

ParentCommandLine "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"

ParentUser UNKNOWN\DEVICE23\csp

Actions

Operational

Open Saved Log...
Create Custom View...
Import Custom View...
Clear Log...
Filter Current Log...
Properties
Disable Log
Find...
Save All Events As...
Attach a Task To This Log...
View
Refresh
Help

Event 1, Sysmon

Event Properties
Attach Task To This Event...
Copy
Save Selected Events...
Refresh
Help

EventID1-Process Creation PowerShell.exe

Event Viewer

File Action View Help

Security-Netlogon
Security-SPP-UX-GenuineCenter-Login
Security-SPP-UX-Notifications
Security-UserConsentVerifier
SecurityMitigationsBroker
SENSE
SenseR
Service Reporting API
SettingsAgentConnector
Shell-ConnectedAccountState
Shell-Core
ShellCommon-StartLayoutPopulation
SmartCard-Audit
SmartCard-DeviceEnum
SmartCard-TPM-VCARD-Module
SmartScreen
SMBClient
SMBDirect
SMBServer
SMBWitnessClient
StateRepository
Storage-Tiering
StorageManagement
StorageManagement-PartUtil
StorageSettings
StorageSpaces-API
StorageSpaces-Driver
StorageSpaces-ManagementAgent
StorageSpaces-Parser
StorageSpaces-SpaceManager
StorageVolume
StorDiag
Store
StorPort
Storsvc
Sudo
Sysmon
SystemSettingsThreshold

Operational Number of events: 244 (0) New events available

Level	Date and Time	Source	Event ID	Task Category
Information	18-06-2026 16:33:20	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	18-06-2026 16:33:20	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	18-06-2026 16:33:20	Sysmon	1	Process Create (rule: ProcessCreate)
Information	18-06-2026 16:33:16	Sysmon	11	File created (rule: FileCreate)
Information	18-06-2026 16:33:15	Sysmon	1	Process Create (rule: ProcessCreate)
Information	18-06-2026 16:33:14	Sysmon	1	Process Create (rule: ProcessCreate)
Information	18-06-2026 16:33:14	Sysmon	1	Process Create (rule: ProcessCreate)

Event 1, Sysmon

General Details

Friendly View XML View

+ System

- EventData

RuleName -

UtcTime 2026-06-18 11:03:20.186

ProcessGuid {07d758bc-d078-6a33-481b-000000000600}

ProcessId 13316

Image C:\Windows\System32\whoami.exe

FileVersion 10.0.26100.1882 (Winbuild.160101.0800)

Description whoami - displays logged on user information

Product Microsoft® Windows® Operating System

Company Microsoft Corporation

OriginalFileName whoami.exe

CommandLine "C:\Windows\system32\whoami.exe"

CurrentDirectory C:\Windows\system32\

User UNKNOWN\DEVICE23\csp

LogonGuid {07d758bc-5489-6a33-d70a-880300000000}

LogonId 0x3880ad7

TerminalSessionId 2

IntegrityLevel Medium

Hashes MD5=956692DADC582CB46E9219F7A5BEFFA,SHA256=23240F9F8B0A8A324110B1C2331DE31DC180E0BF359CB070151A939AF56CD3,IMPHASH=E4464BCD92DF7AC69CCD

ParentProcessGuid {07d758bc-d072-6a33-441b-000000000600}

ParentProcessId 13356

ParentImage C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe

ParentCommandLine "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"

ParentUser UNKNOWN\DEVICE23\csp

Actions

Operational

Open Saved Log...
Create Custom View...
Import Custom View...
Clear Log...
Filter Current Log...
Properties
Disable Log
Find...
Save All Events As...
Attach a Task To This Log...
View
Refresh
Help

Event 1, Sysmon

Event Properties
Attach Task To This Event...
Copy
Save Selected Events...
Refresh
Help

EventID1-Process Creation whoami.exe

Event Viewer

File Action View Help

Security-Netlogon
Security-SPP-UX-GenuineCenter-Loggin
Security-SPP-UX-Notifications
Security-UserConsentVerifier
SecurityMitigationsBroker
SENSE
SenseIR
Service Reporting API
SettingsAgentConnector
Shell-ConnectedAccountState
Shell-Core
ShellCommon-StartLayoutPopulation
SmartCard-Audit
SmartCard-DeviceEnum
SmartCard-TPM-VCARD-Module
SmartScreen
SMBClient
SMBDirect
SMBServer
SMBWitnessClient
StateRepository
Storage-Tiering
StorageManagement
StorageManagement-PartUtil
StorageSettings
StorageSpaces-Api
StorageSpaces-Driver
StorageSpaces-ManagementAgent
StorageSpaces-Parser
StorageSpaces-SpaceManager
StorageVolume
StorDiag
Store
StorPort
StorSvc
Sudo
Sysmon
SystemSettingsThreshold

Operational Number of events: 244 (0) New events available

Level	Date and Time	Source	Event ID	Task Category
Information	18-06-2026 163341	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	18-06-2026 163340	Sysmon	1	Process Create (rule: ProcessCreate)
Information	18-06-2026 163329	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	18-06-2026 163329	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	18-06-2026 163329	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	18-06-2026 163329	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	18-06-2026 163328	Sysmon	1	Process Create (rule: ProcessCreate)

Event 1, Sysmon

General Details

Friendly View XML View

System

EventData

RuleName -

UtcTime 2026-06-18 11:03:28.600

ProcessGuid {07d758bc-d080-6a33-4a1b-000000000000}

ProcessId 16324

Image C:\Windows\System32\HOSTNAME.EXE

FileVersion 10.0.26100.8521 (WinBuild.160101.0800)

Description Hostname APP

Product Microsoft® Windows® Operating System

Company Microsoft Corporation

OriginalFileName hostname.exe

CommandLine "C:\WINDOWS\System32\HOSTNAME.EXE"

CurrentDirectory C:\WINDOWS\System32\

User UNKNOWNWDEVICE23\sp

LogonGuid {07d758bc-5489-6a33-d70a-880300000000}

LogonId 0x3880ad7

TerminalSessionId 2

IntegrityLevel Medium

Hashes MD5=8B2DD25CEB68BA00732F745342C2808SHA256=EE19DA10A0B76B9468BD0626BC1BEFF80B7896CEB878A5A63AF92D6A07E20EEEJMPHASH=BCB84C34505B1E47EF25

ParentProcessGuid {07d758bc-d072-6a33-441b-000000000000}

ParentProcessId 1356

ParentImage C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe

ParentCommandLine "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"

ParentUser UNKNOWNWDEVICE23\sp

Actions

Operational

Open Saved Log...
Create Custom View...
Import Custom View...
Clear Log...
Filter Current Log...
Properties
Disable Log
Find...
Save All Events As...
Attach a Task To This Log...
View
Refresh
Help

Event 1, Sysmon

Event Properties
Attach Task To This Event...
Copy
Save Selected Events...
Refresh
Help

EventID1-Process Creation hostname.exe

Event Viewer

File Action View Help

Security-Netlogon
Security-SPP-UX-GenuineCenter-Loggin
Security-SPP-UX-Notifications
Security-UserConsentVerifier
SecurityMitigationsBroker
SENSE
SenseIR
Service Reporting API
SettingsAgentConnector
Shell-ConnectedAccountState
Shell-Core
ShellCommon-StartLayoutPopulation
SmartCard-Audit
SmartCard-DeviceEnum
SmartCard-TPM-VCARD-Module
SmartScreen
SMBClient
SMBDirect
SMBServer
SMBWitnessClient
StateRepository
Storage-Tiering
StorageManagement
StorageManagement-PartUtil
StorageSettings
StorageSpaces-Api
StorageSpaces-Driver
StorageSpaces-ManagementAgent
StorageSpaces-Parser
StorageSpaces-SpaceManager
StorageVolume
StorDiag
Store
StorPort
StorSvc
Sudo
Sysmon
SystemSettingsThreshold

Operational Number of events: 244

Level	Date and Time	Source	Event ID	Task Category
Information	18-06-2026 163341	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	18-06-2026 163341	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	18-06-2026 163341	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	18-06-2026 163341	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	18-06-2026 163340	Sysmon	1	Process Create (rule: ProcessCreate)
Information	18-06-2026 163329	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	18-06-2026 163329	Sysmon	13	Registry value set (rule: RegistryEvent)

Event 1, Sysmon

General Details

Friendly View XML View

System

EventData

RuleName -

UtcTime 2026-06-18 11:03:40.850

ProcessGuid {07d758bc-d08c-6a33-4b1b-000000000000}

ProcessId 11044

Image C:\Windows\System32\ipconfig.exe

FileVersion 10.0.26100.8521 (WinBuild.160101.0800)

Description IP Configuration Utility

Product Microsoft® Windows® Operating System

Company Microsoft Corporation

OriginalFileName ipconfig.exe

CommandLine "C:\WINDOWS\System32\ipconfig.exe"

CurrentDirectory C:\WINDOWS\System32\

User UNKNOWNWDEVICE23\sp

LogonGuid {07d758bc-5489-6a33-d70a-880300000000}

LogonId 0x3880ad7

TerminalSessionId 2

IntegrityLevel Medium

Hashes MD5=E4B180282B5F3056414B0C6CCD5238E2SHA256=2AF2AB72B6DCA3B5B6F4C3B05C9284ACBDF0DBB84502D3E47BD338C62C26DA6JMPHASH=AC16E1EA98679DC4FA6

ParentProcessGuid {07d758bc-d072-6a33-441b-000000000000}

ParentProcessId 1356

ParentImage C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe

ParentCommandLine "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"

ParentUser UNKNOWNWDEVICE23\sp

Actions

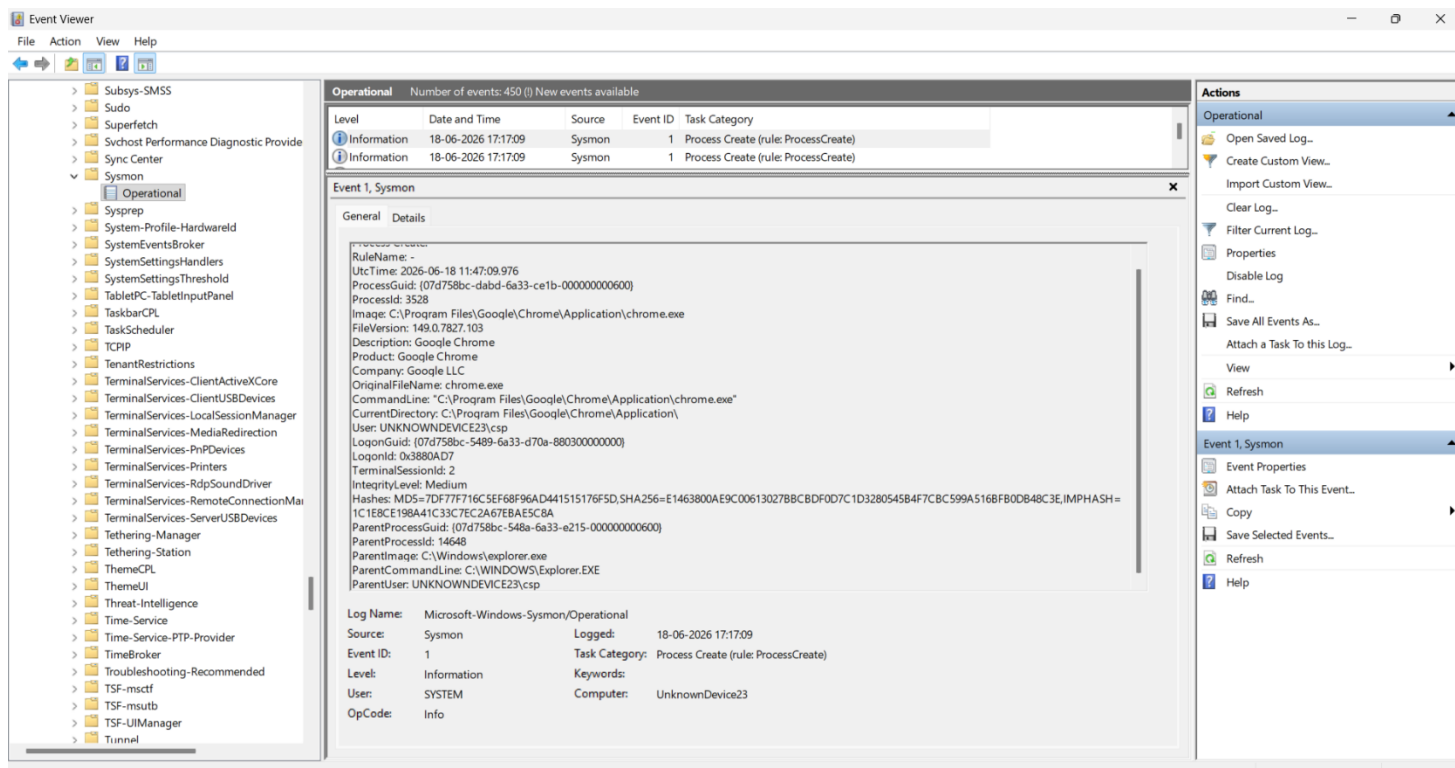
Operational

Open Saved Log...
Create Custom View...
Import Custom View...
Clear Log...
Filter Current Log...
Properties
Disable Log
Find...
Save All Events As...
Attach a Task To This Log...
View
Refresh
Help

Event 1, Sysmon

Event Properties
Attach Task To This Event...
Copy
Save Selected Events...
Refresh
Help

EventID1-Process Creation ipconfig.exe



EventID1-Process Creation chrome.exe

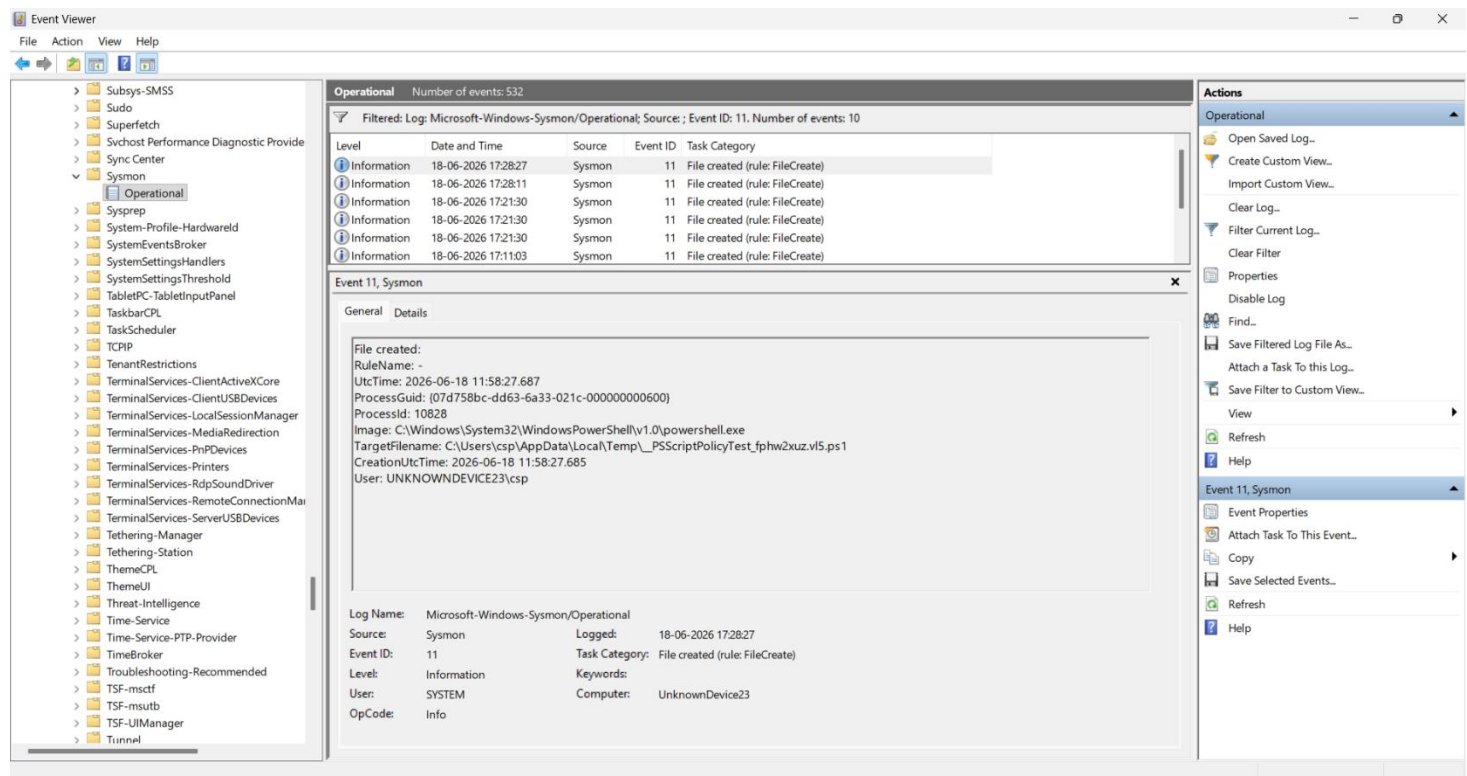
5. Event ID 11 - File Creation

Activities performed:

- Created test files using PowerShell and Notepad.

Findings:

- Identified file creation events.
- Tracked the process responsible for creating files.



EventID11-File Creation notepad.exe and PowerShell

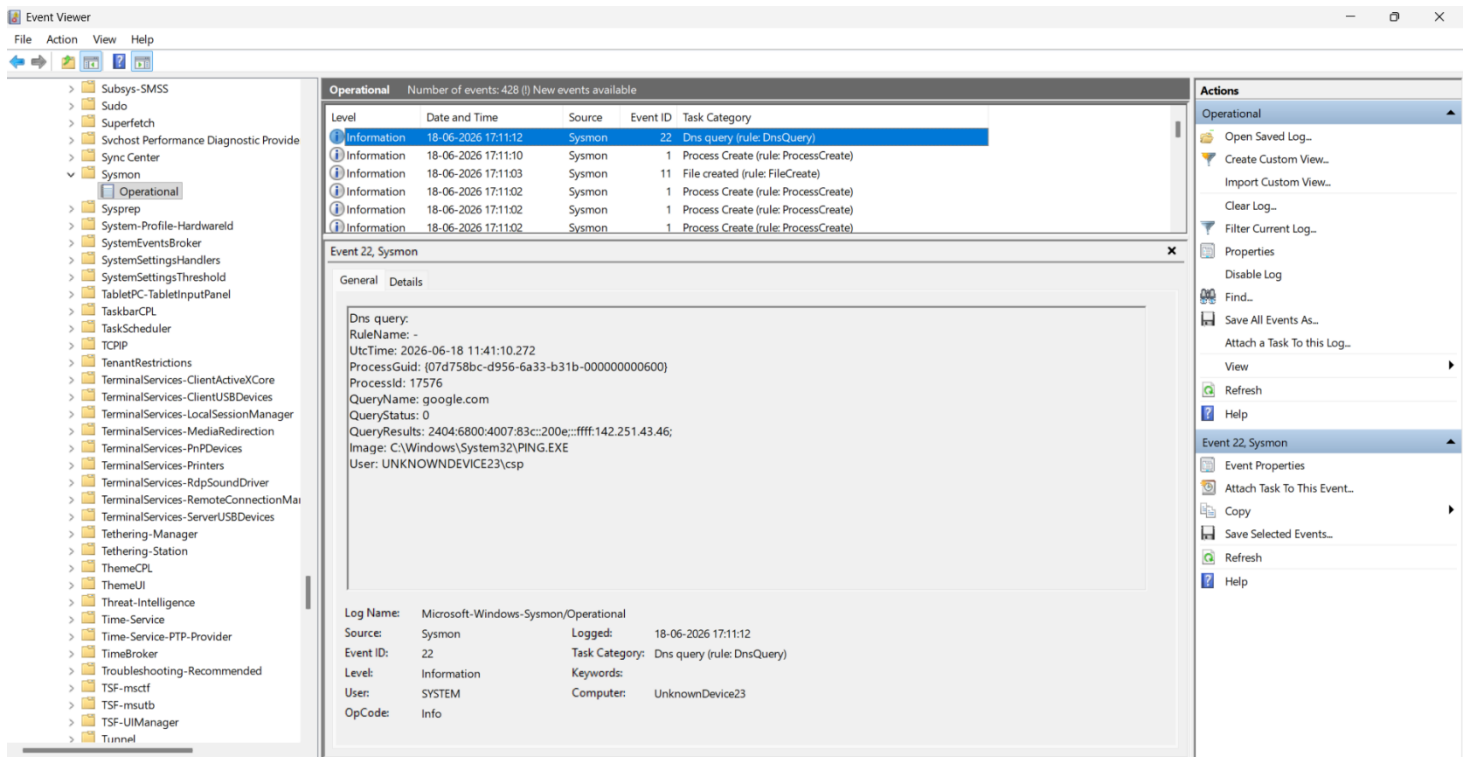
6. Event ID 22 - DNS Query

Activities performed:

- Accessed google.com.

Findings:

- Observed DNS query events.
- Identified the process generating DNS requests.



- EventID22-DNS Query Google.com

7. Conclusion

This project successfully demonstrated Windows endpoint monitoring using Sysmon and Event Viewer. The lab provided hands-on experience with process creation, file creation, and DNS query analysis, which are important skills for SOC Analysts.

Author

Mohan R
SOC Analyst Home Lab Project

